

40 / 80 column switcher for CBM 8032080 and 8032089 main boards

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Historical background

Later in the life of the 80 column 8032, Commodore switched the 40 column models to have the same 12 inch CRT display. At the same time, the main board for the 8032 was made universal, by introducing a set of jumper wires to either use the main board for 40 or 80 column operation.

This in turn means that such a main board of an 80 column machine can be converted to a 40 column one, and vice versa. A step further is to add a switching facility to switch between 40 and 80 column operation. This project provides a possible solution to achieve this.

Please note that the first revision of the main board for the 8032 (board 8032030) cannot be converted to 40 column operation using the switch presented here. The later production of the models 8096 and SP9000/SuperPET use compatible main boards, but the switch presented here will mechanically interfere with the extra boards in these computers.

Main board differences between 40 and 80 column models

Because the 80 column screen requires twice the amount of video RAM, two extra 2114 RAM chips and three extra logic devices are present on the main board. They are located on the following locations:

- UB6: 74LS244 octal bus buffer, data bits 0-3;
- UB7: 74LS244 octal bus buffer, data bits 4-7;
- UB8: 74LS373 octal latch;
- UC6: 2114 RAM, data bits 0-3;
- UC7: 2114 RAM, data bits 4-7.

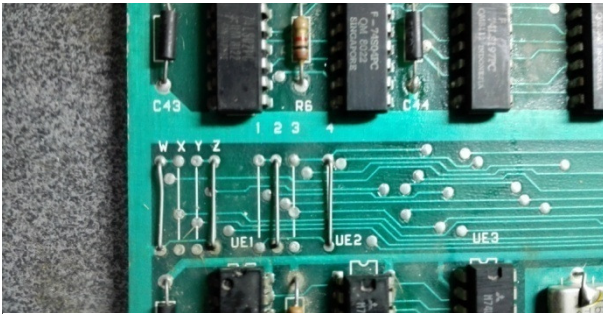
On 40 column models these locations on the main board are not populated.

Jumper fields

To facilitate conversion from 40 to 80 columns and vice versa, three jumper fields are present:

- J1 through J4. These switch the clock signals;
- J5 through J8, These switch the enables for the buffers in the video RAM circuitry;
- 10 Jumpers to move the connection between the CPU address lines and the video RAM address lines one bit up or down.

On an 8032080 main board, the jumper field J1-J4 is located near the right front of the main board and looks like this:



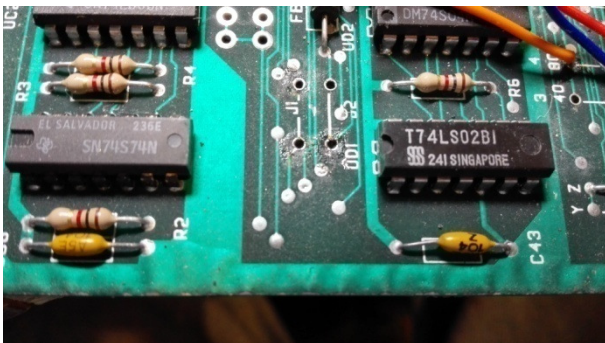
40 column:
J1, J3 closed; J2, J4 open

80 column:
J1, J3 open; J2, J4 closed (shown)

Note: jumpers W-Z are for the RAM configuration.

Figure 1: Main board 8032080, J1-J4 jumpered for 80 column

Main board 8032089 has the J1/J2 fields in a different location and it looks like this:

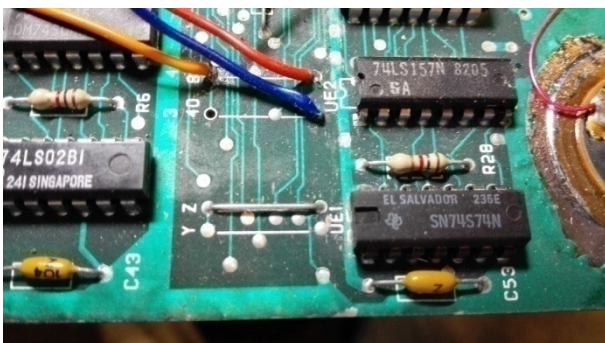


40 column:
J1 closed; J2 open

80 column:
J1 open; J2 closed

Figure 2: Jumper field J1/J2 on a 8032089 main board

The J3/J4 jumper field on main board 8032089 looks like this:



40 column:
J3 closed; J4 open

80 column:
J3 open; J4 closed

Figure 3: Jumper field J3/J4 on a 8032089 main board

The jumper field J5-J8 is located about one third from the left, and further inwards. It looks like this:

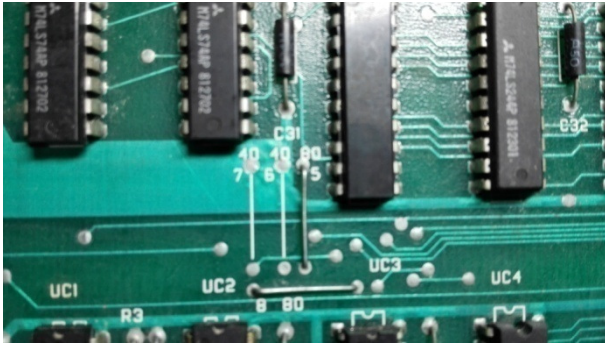


Figure 4: Main board 8032080, jumpered for 80 column
(8032089 main board is effectively the same)

40 column:
J6, J7 closed; J5, J8 open

80 column:
J6, J7 open; J5, J8 closed (shown)

The jumper field to move the address lines is located in the centre of the main board and looks like this:

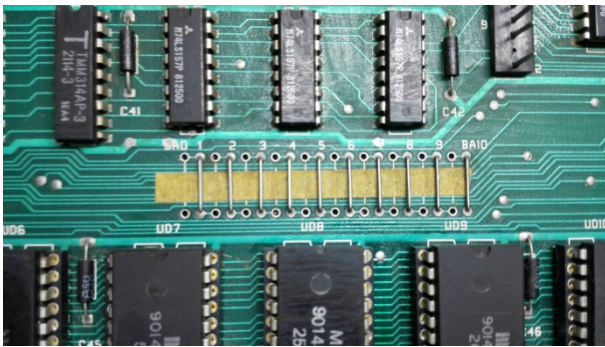


Figure 5: Main board 8032080, jumpered for 80 column
(8032089 main board uses a slightly different layout)

40 column:
Every odd jumper in, every even jumper out
(so BA0 in, BA10 out)

80 column:
Every even jumper in, every odd jumper out
(so BA0 out, BA10 in (shown))

Switching between 40 and 80 column mode

To switch between 40 and 80 column mode, the following must be changed on the main board:

- The 10 video RAM address lines must be switched. This can be done using 74LS157 or 74LS257 devices. As 10 lines are to be switched, this will require 2.5 74LS157/257 ICs (each IC contains 4 switches). This part handles the change in address lines for the video RAM;
- The dot clock must be changed between 8 and 16 MHz. This is handled by the jumpers J1 and J2;
- The LOADSR signal must be modified. This is handled by the jumpers J3 and J4 on the main board. These have one common connection, making this a SPDT configuration that can be handled by a section of a 74LS157/257;
- The CLK of the latch for the even addresses at UB5 (74LS373) must be switched between CLK1 and GND. This is handled by J5 and J6. These have common connection as well, making this a SPDT configuration that can be handled by a section of a 74LS157/257;
- The signals /WRITE_EVEN and /READ_EVEN need to be switched between controlled by BA0 or always active. This is handled by J7 and J8 on the main board. Again these have a common connection (making this a SPDT configuration) that can be handled by a section of a 74LS157/257;
- In 40 column mode, the latch at UB8 must be disabled to switch out the odd display RAM section. This too can be achieved using a section of a 74LS157/257.

In total, 14 jumpers and one connection are to be handled, requiring four 74LS157/257 devices; one section is not used.

Schematic diagram

The schematic diagram of the entire 40/80 column switch adapter is given below.

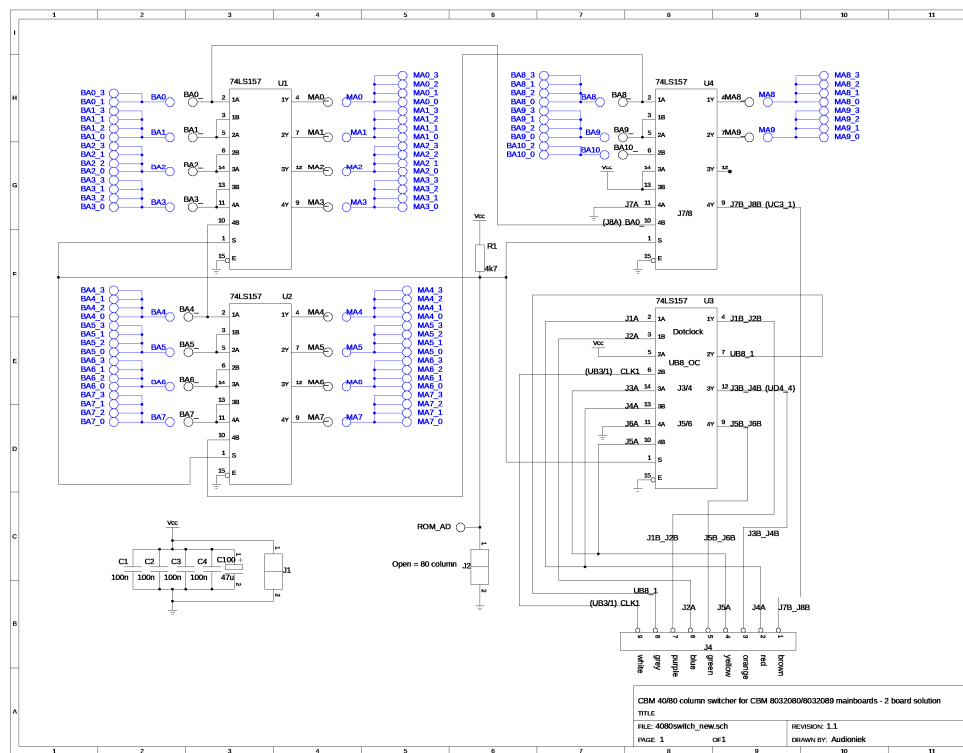


Figure 6: Schematic diagram

The parts shown in blue are on the smaller PCB, the black parts are located on the larger PCB of the switch circuitry.

All address line connections in blue (BAX_Y) are given twice or four times in order to provide a pin out for both the 8032080 and 8032089 main boards, as these boards use a different lay out.

U1, U2 and half of U4 perform the address line switching. One switch in U4 and all in U3 take care of the removed jumpers on the main board and disabling UB8 in 40 column mode. These switches are used as follows:

- J1/J2: U3, pins 2, 3 and 4 (dot clock 8 or 16 MHz);
- J3/J4: U3, pins 12, 13 and 14 (/LOADSR signal);
- J5/J6: U3, pins 9, 10 and 11 (enable even data latch);
- J7/J8: U4, pins 9, 10 and 11 (BA0 switch).

The odd data latch at location UB8 must be disabled in 40 column mode. U3, pins 5, 6 and 7 take care of this. U4, pins 12, 13 and 14 are not used; the inputs are tied to Vcc.

The various connectors on the switch have these functions:

J1 is the connector for the supply voltage.

J2 is the connector for the switch to select 40 or 80 column mode (connected is 40 columns).

J3 is the output for the extra address line for the editor-ROM in order to select the correct half depending on the column mode selected (low is 40 columns).

J4 provides the interface to the main board regarding all non-address line connections. 9 extra wires are needed:

- Pin 1: connected to UC3 pin 1 (common of J7 and J8);
- Pin 2: connected to UD3 pin 5 (J4A);
- Pin 3: connected to UD4 pin 4 (common of J3 and J4);
- Pin 5: connected to UE4 pin 10 (common of J5 and J6);
- Pin 6: connected to UD2 pin 10 (J2A);
- Pin 7: connected to UC1 pin 11 (common of J1 and J2);
- Pin 8: connected to UB8 pin 1 (UB8, pin 1);
- Pin 9: connected to UC8 pin 1 (UC8, pin 1).

Modifying the main board

Now it is time to remove the main board from the computer. Proceed as follows:

- Unplug the computer from the mains supply;
- Disconnect all peripherals from the computer;
- Remove the screws attaching the front of the top case to the bottom chassis;
- Lift the top case up like you would the hood/bonnet of a car and use the rod provided in the bottom to support it in the open position;
- Locate the following connectors on the main board and note their orientation. Take photographs if you are not sure on how to put them back, although each connector is different and keyed to prevent wrong connection. There are three of them:
 - Middle, somewhat to the right (8032080 board), or left front (8032089 board), small: video;
 - Left front, large: power;
 - Middle, large: keyboard.
- Remove the connectors from the main board;
- The main board is attached to the bottom chassis with three screws and three plastic standoffs. Remove the three screws (two at the corners on the rear, one at the middle right);
- Start with right front corner. Press in the little notch on the standoff while at the same time trying to lift the main board. As the main board comes loose, lift it a little and let it rest on the standoff. Do the same with the standoff on the left corner, and finish with the one in the left middle;
- Take the main board out of the computer.

Using a desoldering tool, remove the four bridge wires that determine the column mode and leave the holes for them open. On a 40 column model, the wires J1, J3, J5 and J7 will be present, on an 80 column one J2, J4, J6 and J8. On an 8032080 main board leave J2 in if present; add it if absent.

Also remove the 10 jumper wires in the address jumper field using the same method. Again, leave the jumper holes open.

There is one modification to be made on the main board, irrespective of the version. The trace to device UB8 pin 1 on the solder side must be cut near the pin. Turn the main board over and locate this spot:

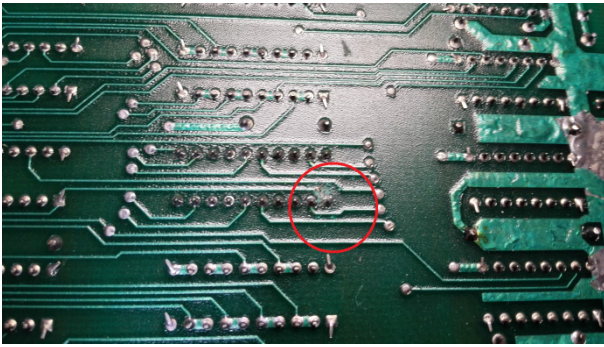


Figure 7: Cut trace to UB8 pin 1.

Cut the trace shown on the photo above near pin 1 of UB8 pin1. The photo shows the trace already cut.

Adding missing ICs

If you are converting a 40 column machine to 80 columns, five missing ICs must be added. Consult the bill of materials in appendix A which device goes where. The use of IC sockets is recommended.

If the ICs are indeed not present on the main board, first open the solder islands for them by using a desoldering tool. Then solder in the ICs sockets (or the ICs directly if you want to). Place the ICs in their sockets if using the latter. If capacitors C38 and C39 are missing as well, add them, using 100nF ceramic capacitors.

Modifying the 8032080 main board

On the 8032080 main board, a design error is present that may or may not have been corrected when the board or computer was manufactured. When the board is wired for 80 column video the error is not significant, but in 40 column mode it is. It can generally be stated that a main board in a 40 column machine has been corrected but one in an 80 column machine usually is not.

To make matters worse, the original Commodore drawing of the 8 or 16 MHz clock circuit is drawn incorrectly. The original schematic is drawn like this:

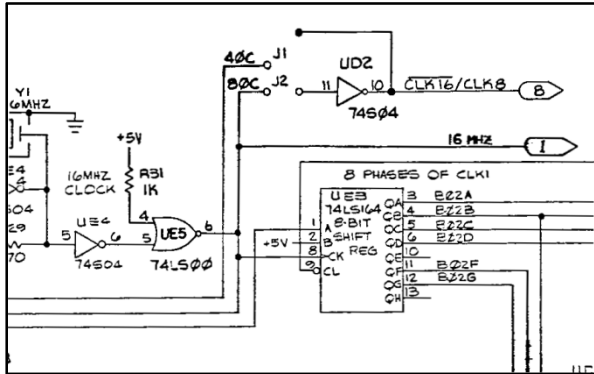


Figure 8: Incorrect schematic of 8032080 clocks

In 80 column mode, J2 would be closed, connecting the 16MHz clock from UE5 pin 6 via the inverter UD2 (pins 10 and 11) to the video circuitry.

In 40 column mode an 8 MHz clock is connected through J1. This is cannot be correct, as this connects the output of inverter UD2 to the output of the clock divider, causing a conflict.

The circuit actually present on the PCB is different, and its schematic looks like this:

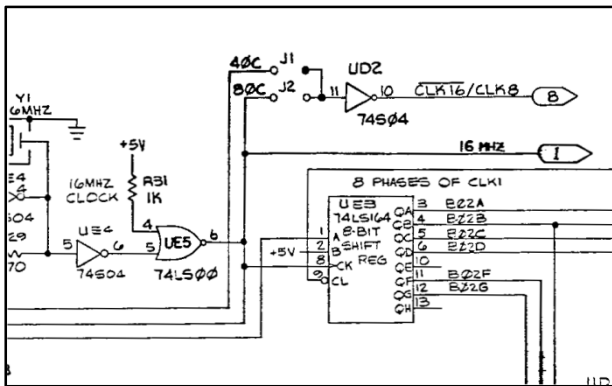


Figure 9: The circuit actually present on the 8032080 main board

Although the above mentioned conflict does not exist in the actual circuit, the circuit will work in 80 column mode, but not in 40 column mode, because the 8 MHz clock is inverted through the UD2 inverter when J1 is closed.

The circuit we need is this:

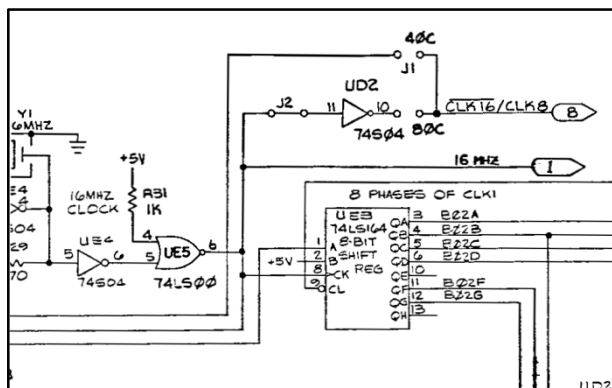


Figure 10: Modified clock circuit for the 8032080 main board

Jumper 2 is left in (or is connected) in all cases, so pin 11 of inverter UD2 receives the 16 MHz clock. The connection between pin 10 of UD2 and the signal /CLK16 / CLK8 is cut. The clock for 40 column mode can be taken from the correct island for J1.

The modification of main board 8032080 involves the following.

- If present, remove the bridge wire J1;
- If absent, add a bridge wire for J2;
- Isolate pin 10 of IC UD3 (74S04) as follows.
 - Carefully study the photograph below (it is of the area immediately above IC UC2) and identify the via circled in red in the photograph (it has the blue wire connected to it in the photograph);



- On the solder side, the via is connected to pin 10 of UD3 and will be used for a connection to the switch PCB. On the component side a trace runs from the via towards the front of the main board where it is connected to UC1 pins 3 and 11. If in doubt about the correct via, check for continuity between the via and the IC pins just mentioned. The trace is hard to see, as it is covered by resistor R5. Carefully cut the trace just above the resistor (indicated by the red arrow in the photograph above), leaving the via untouched. There should be no continuity between the via and pins 3 and 11 of UC1 but continuity between IC UD3 pin 10 and the via now.
- This completes the modification of the 8032080 main board.

Building the adapter

The PCBs for the adapter come as one board. Split it in two along the silk line separating the two with a saw or Dremel tool:

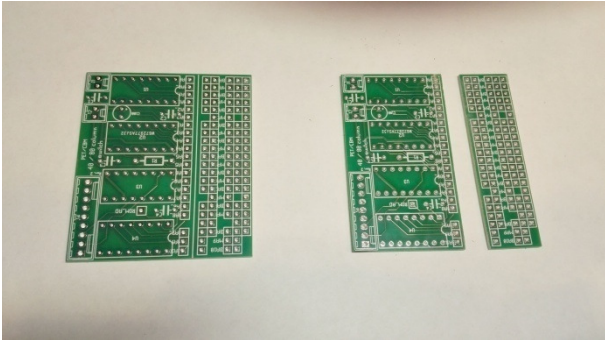


Figure 11: Split PCB

Building the larger PCB

Solder the angled pin headers to the larger PCB. You need a 3 pole and an 18 pole header:

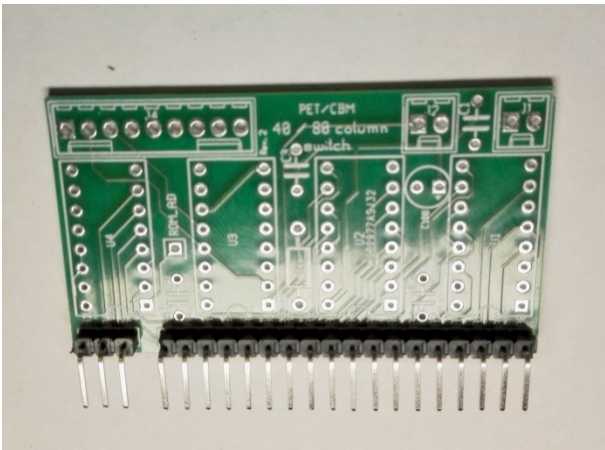


Figure 12: Large PCB with angled connector pins

As an alternative, use a strip of 22 contacts and remove the pin not needed. Warming the pin with a soldering iron will make removing easier.

Mount the other components on the larger PCB. When finished it should look like this:

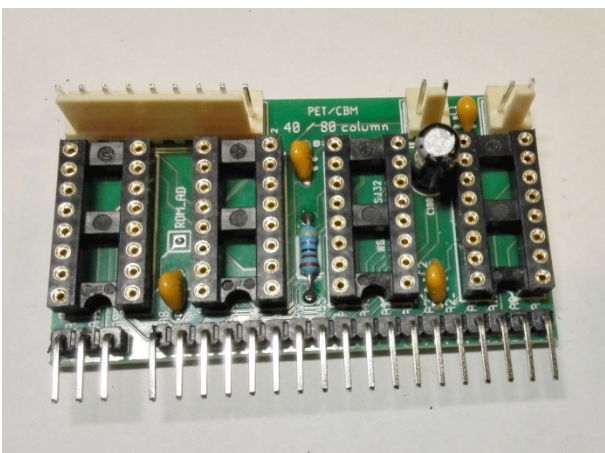


Figure 13: Completed large PCB, sans ICs

The use of IC sockets and the KF2510 connectors is optional but recommended. If you use sockets, use good quality ones; cheaper sockets cause more problems than they solve in the long run.

Insert or solder the four ICs. Either 74LS157 or 74LS257 types will work, they can even be mixed.

Adding the address connectors to the main PCB of the computer

Decide whether you want female receptacles or pins on the main board (the switch PCB assembly will of course have the opposite gender of connectors). Either will work. Prepare a set of connectors; they are different for the 8032080 and 8032089 main boards:

- 8032080 main board: two rows of 20 pins/receptacles;
- 8032089 main board: two rows of 5, two of 6 and two of 9 pins/receptacles each.

Note: the female receptacles are usually sold in strips of 40 contacts. In theory, you need one strip of 40. It is however rather difficult to exactly cut a strip between two contacts, without losing a contact. Prepare for some loss of contacts or try to buy ready made strips with the number of contacts needed if possible.

Solder the pins/receptacles into the address jumper field(s) on the main board. First solder one or both end contacts/pins, then check the pins/receptacle(s) are upright. Correct if needed, then solder the remaining contacts. The result should look like this:

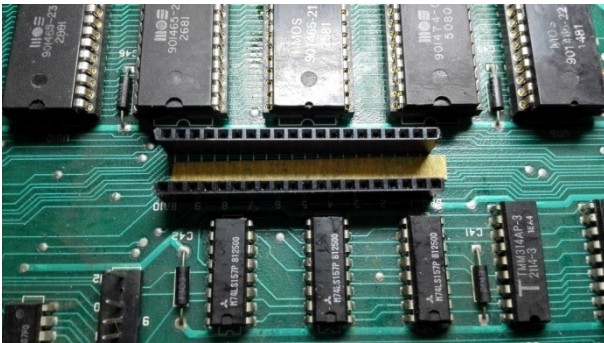


Figure 14: 8032080 main board with female receptacles

Next, insert the corresponding pins/receptacles on the connectors in the main board:

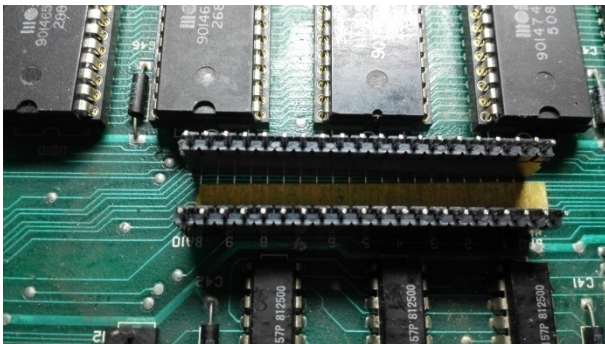


Figure 15: Pins inserted in female receptacles (8032080 board)

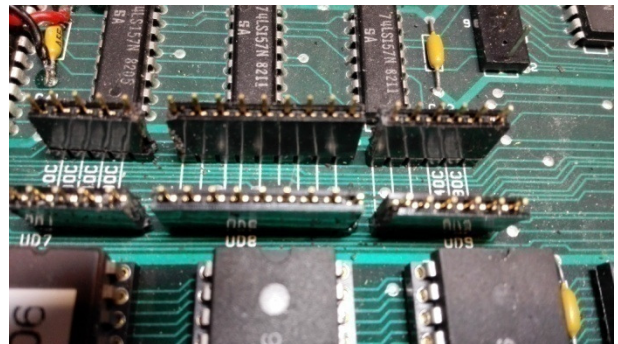
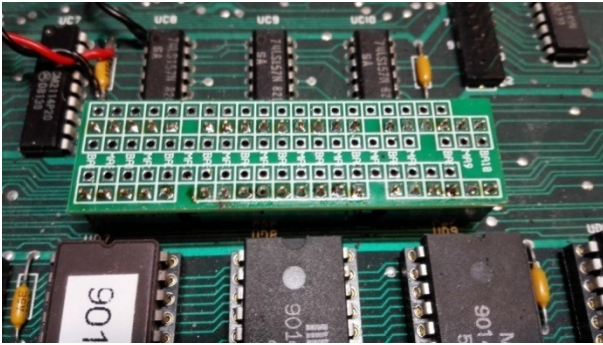


Figure 16: Receptacles attached to pins in main board (8032089 board)

Put the small PCB on top of the pins/receptacles. On a 8032089 main board this will fit only one way; on a 8032080 main board the orientation should be that the white silk screen print should be on top and the missing centre pin should nearest the rear side of the main board.

Solder the pins/receptacles to the small PCB:



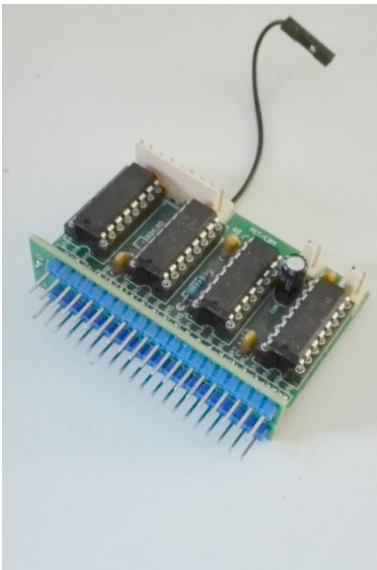
Note: 8032089 main board shown, large PCB not mounted yet.

Figure 17: Soldered small PCB in 8032089 main board

Take the assembly of the small PCB out of the main board.

Attaching the small PCB

The small PCB is connected to the large one through the centre row of holes. Due to the missing pin in the row, it only fits in one way as the connectors for the main board should obviously be facing down: the side with the silk screen text should point to the large PCB. Align the pins of the large PCB with the holes in the small one. Solder one end pin and check the small PCB is perpendicular with the large one. Correct if needed. Solder the other end pin, and recheck the boards are perpendicular and straight. Correct if needed. Then solder the remaining pins. Finish off by cutting off the excess length of the centre pins on the solder side of the small PCB. The result should look like this (version for 8032080 main board):



The black wire shown is the connection for A11 of the editor ROM and is connected to J3 (ROM_AD).

Figure 18: The complete switch

Insert the switch assembly into the main board again. The components should point to the side with the power transformer (in case of board 8032089: it fits in only one way if all pins connect).

Preparing the other connections to the main board

Prepare the remaining connections to the main board as follows (the text assumes the KF2510 connectors are used). The wire colours mentioned are recommended ones and are used throughout the remaining text.

First prepare the loom for J1 (power connection) using red and black wires. The red wire goes in pin 1 of the connector housing (the rearmost when connected to the switch assembly). The black wire goes in pin 2.

Attach the 2-pin KF2510 housing to the connector J1 of the switch assembly. The red wire is connected to the leftmost side of C41 on the main board; the black wire to the other side of the same capacitor. Cut the wires to length, strip them and solder them to capacitor C41. The power connection is now complete.

Attach wires to the 9 pin KF2510 housing as follows:

Pin	Colour
1	Brown
2	Red
3	Orange
4	Yellow
5	Green
6	Blue
7	Purple
8	Grey
9	White

Note: the colours mentioned match the colour coding used on resistors and are a suggestion; they are used throughout the following text. The colour is of course irrelevant; as long as the correct pin of J4 is connected to the correct main board point, everything should work.

Remove the switch assembly from the main board and attach the loom for J4 to it. If you prefer, bind the wires together using pieces of shrink tubing of 2 cm length about 2 cm from the connector housing and again about 6 cm from the housing. Then fold the wires under the switch assembly between the protruding pins/receptacles and mount the switch assembly on the main board again. The result should look like this:

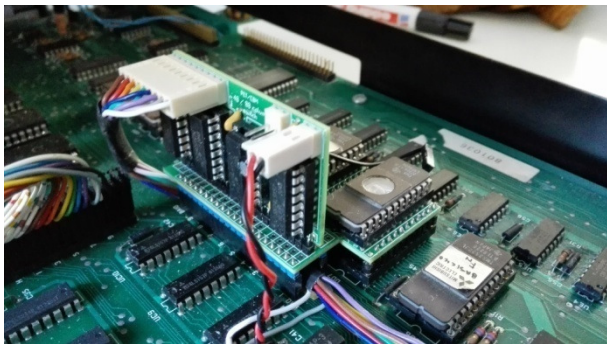


Figure 19: The switch in place, power connected

Now the connections to the main board can be made. Depending on the main board type, these are somewhat different.

Wiring the 8032080 main board

Between the address connectors on the main board, nine wires emanate. Connect these to the following points on the main board:

KF2510 pin#	Wire colour	Main board location	Remark
1	Brown	J8, front side	Also see photograph below
2	Red	J4, right side	Also see photograph below
3	Orange	J3 or J4, left side	Also see photograph below
4	Yellow	J5, right side	Also see photograph below
5	Green	J6, right side	Also see photograph below
6	Blue	Via between UD2 and UD3	Also see photograph below
7	Purple	UC1 (74S74), pin 11	
8	Grey	UB8 (74LS373), pin 1	
9	White	UC8 (74LS157), pin 1	

All connections are made on the component side of the main board. Cut the wire to length, strip it, and solder to the point indicated in the table above.

Detailed wiring photographs(8032080 main board):

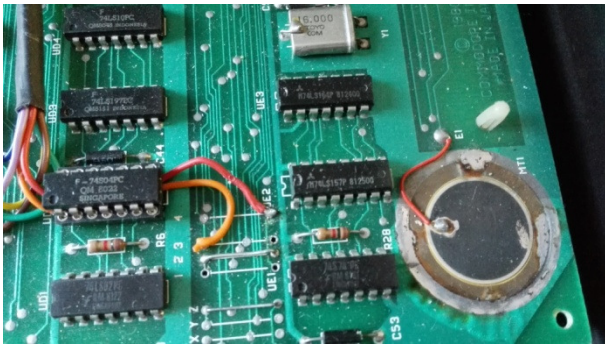


Figure 20: Detail of J3/J4 (note: J2 inserted)

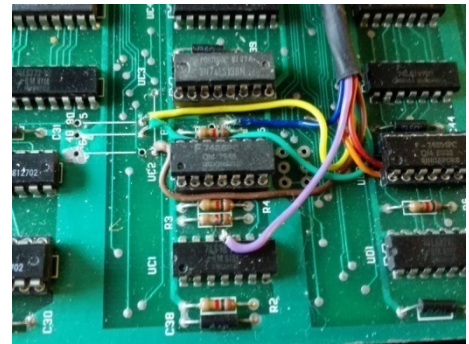


Figure 21: Detail of J5/J6 and clock wiring

Wiring the 8032089 main board

Between the address connectors on the main board, nine wires emanate. Connect these to the following points on the main board:

KF2510 pin#	Wire colour	Main board location	Remark
1	Brown	J8, front side	Also see photograph below
2	Red	J4, right side	Also see photograph below
3	Orange	J3 or J4, left side	Also see photograph below
4	Yellow	J5, right side	Also see photograph below
5	Green	J6, right side	Also see photograph below
6	Blue	J2 front	Also see photograph below
7	Purple	J2 rear	Also see photograph below
8	Grey	UB8 (74LS373), pin 1	
9	White	UC8 (74LS157), pin 1	

All connections are made on the component side of the main board. Cut the wire to length, strip it, and solder to the point indicated in the table above.

Detailed wiring photographs(8032089 main board):

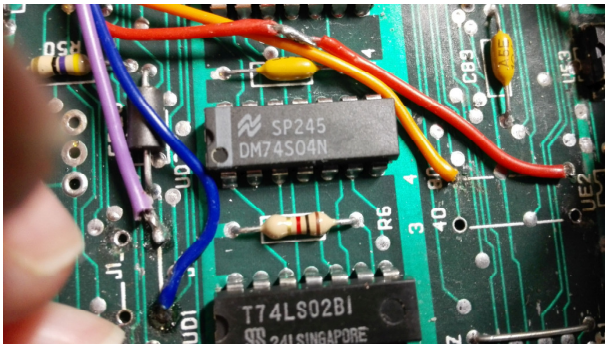


Figure 22: Detail of J1/J2 and J3/J4

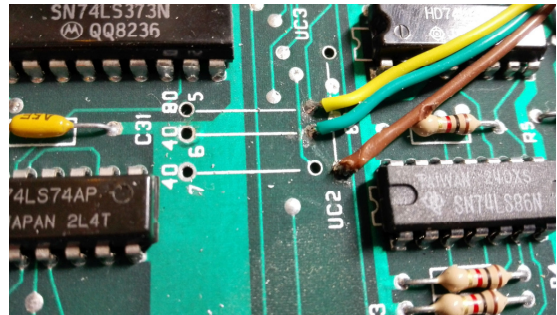


Figure 23: Detail of J5/J6 and J7/J8

The editor ROM

The editor ROM (address range \$E000-\$E7FF) is different for 40 and 80 column modes and the problem is that the full set needed have not been made by Commodore in the past. Per BASIC version, the following ROMs are needed, as there are two keyboard lay outs, two CRT refresh rates and two column widths to deal with.

BASIC 4.0

Keyboard	CRT refresh	Columns	ROM#
N	50 Hz	40	901498-01
N	60 Hz	40	901499-01
N	50 Hz	80	Not available originally
N	60 Hz	80	Not available originally
B	50 Hz	40	Not available originally
B	60 Hz	40	Not available originally
B	50 Hz	80	901474-04
B	60 Hz	80	901474-03

For BASIC4.0 it can be seen that no ready made editor ROM was made by Commodore for 40 column, B-keyboard, and 80 column, N-keyboard. The latter case has been addressed in the past by several people, usually by changing an editor ROM for the 8032 to have the N-keyboard layout, powering up in graphics mode and catering for the absence of the REPEAT and ESC keys. Such a ROM, including its source code, is available in the git. Along the same way the 901498-01 and 901499-01 ROMs have been given the same treatment by replacing the keyboard decoding table, changing the power up mode to text, and adding the different processing for shifted characters. (Commodore forgot to remove the support for the REPEAT key.) Again, both the binary and the source code are available in the git.

With regard to the older BASIC versions (1.0 and 2.0), Commodore did not make any editor ROM that can be used. For 80 column mode none were made as the 8032 did not exist yet, and all 40 column editor ROMs do not initialize the CRTIC chip. In the future, this will be addressed in the Super-ROM git.

For an alternative, see <https://github.com/sjgray/cbm-edit-rom/tree/master>. Please note that this git only provides for BASIC 4.0.

The switch assembly switches everything on the main board to facilitate 40 or 80 column operation, including switching the editor ROM needed. For this a single solder island marked ROM_AD is provided. This connection carries a logic high in 80 column mode and a logic low in 40 column mode. Depending on the situation in hand there are several ways to achieve switching the editor ROM. The description below is possibly the simplest one.

If you have access to a 2532 EPROM and the facility to program it, program the EPROM with the ROM images provided in the git you need (N- or B-keyboard, 50 or 60 Hz CRT refresh) and insert it in place of the editor ROM currently on the main board. The ROM image for 40 column goes in the bottom half of the EPROM; that for 80 column in the upper half. The editor ROM (location UD7) is usually the only ROM in a socket. Before inserting, bend pin 18 of the EPROM slightly outward so that it is not in the socket on the main board. Connect pin 18 of the EPROM to the island ROM_AD on the switch assembly.

In case you want to use a 2732(A) EPROM, program it with the ROM versions needed, and use a 2532->2732 adapter that brings out address line A11 of the 2732 and leaves pin 18 of the ROM socket unconnected. Connect the A11 connection of the adapter to the island ROM_AD on the switch assembly.

For other scenarios and needs, all individual ROM images for BASIC 4.0 have been provided in the git as well.

Appendix A: bills of material

The BOM for the missing ICs on the main board in case of a 40 column machine

Location	Description	Qty	Notes
UC6, UC7	2114 1kx4 static RAM	2	350 ns or faster
UB6, UB7	74LS244	2	
UB8	74LS373	1	
UC6, UC7	IC socket, 300 mil wide, 18 pin	2	Optional
UB6, UB7, UB8	IC socket, 300 mil wide, 20 pin	3	Optional

The BOM for the switcher

Location	Description	Qty	Notes
R1	Resistor, 0207, 4k7ohm	1	
U1, U2, U3, U4	74LS157 or 74LS257	4	
C1, C2, C3, C4	Capacitor, ceramic, 100nF, 200 mil, radial	4	
C100	Capacitor, electrolytic, 47µF, 10V or more, radial, 100 mil	1	
-	PC board		
J1, J2	Molex KF2510 male, upright, 2 pin	2	Optional
J4	Molex KF2510 male, upright, 9 pin	1	Optional
-	Molex KF2510 female, 2 pin	2	Optional
-	Molex KF2510 female, 9 pin	1	Optional
-	Molex KF2510 crimp contact	13	Optional
-	Pin strip, male, 40 pins in total		See note 1
-	Pin strip, male, angled 21 or 22 pins in total		See note 3
-	Pin receptacle, female, 40 pins in total		Optional, see note 2
-	IC socket, 300 wide, 16 pin	4	Optional

Optional items: the use of connectors and IC sockets is strongly recommended but not strictly necessary. Using them gives the possibility to easily take things apart again and make problem solving easier.

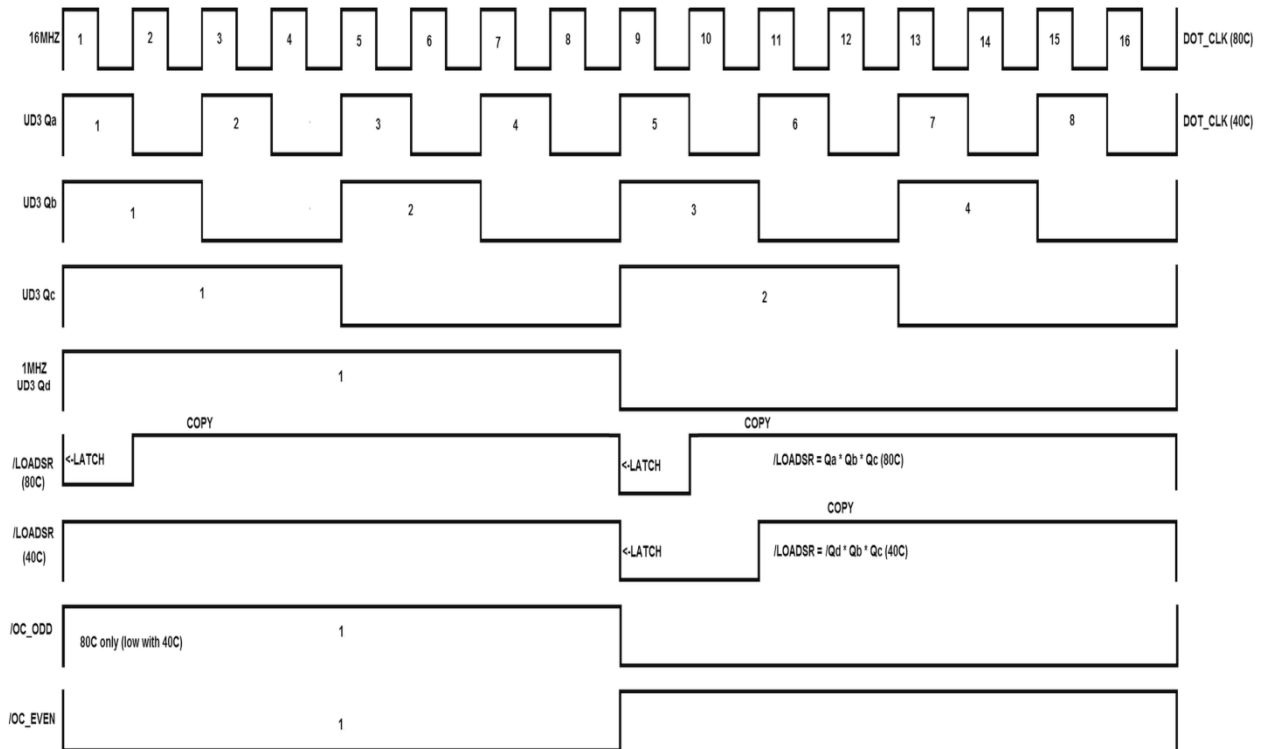
Note 1: for main board 8032080 you need two strips of 20 pins, for main board 8032089 you need two strips of 5 pins, two of 6 pins and two of 9 pins. In each case it is easiest to buy one strip of 40 pins and cut it as required.

Note 2: for main board 8032080 you need two strips of 20 receptacles, for main board 8032089 you need two strips of 5 pins, two of 6 pins and two of 9 pins. Although it is possible to create these out of one strip of 40 receptacles, it is very hard to cut a larger strip without losing a contact in the process, although it can be done. If you can get strips in the pin counts required directly, this is recommended. Making the/cutting the strips yourself can be done starting with one strip of 40 receptacles, but be prepared to lose some contacts, so buy more receptacles than you need in this case, just to be sure.

Note 3: 21 pins are required. Cut three pins off a strip of 21, or remove the pin not needed off a strip of 22. See the photographs for details.

Appendix B: timing diagram

Below an overview is given of the various signals involved in the switching of the column width. The reference is a sequence of 16 clock periods of the 16MHz clock, out of which all other signals required are generated.



For more details and insight, consult a schematic diagram of the main board.

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